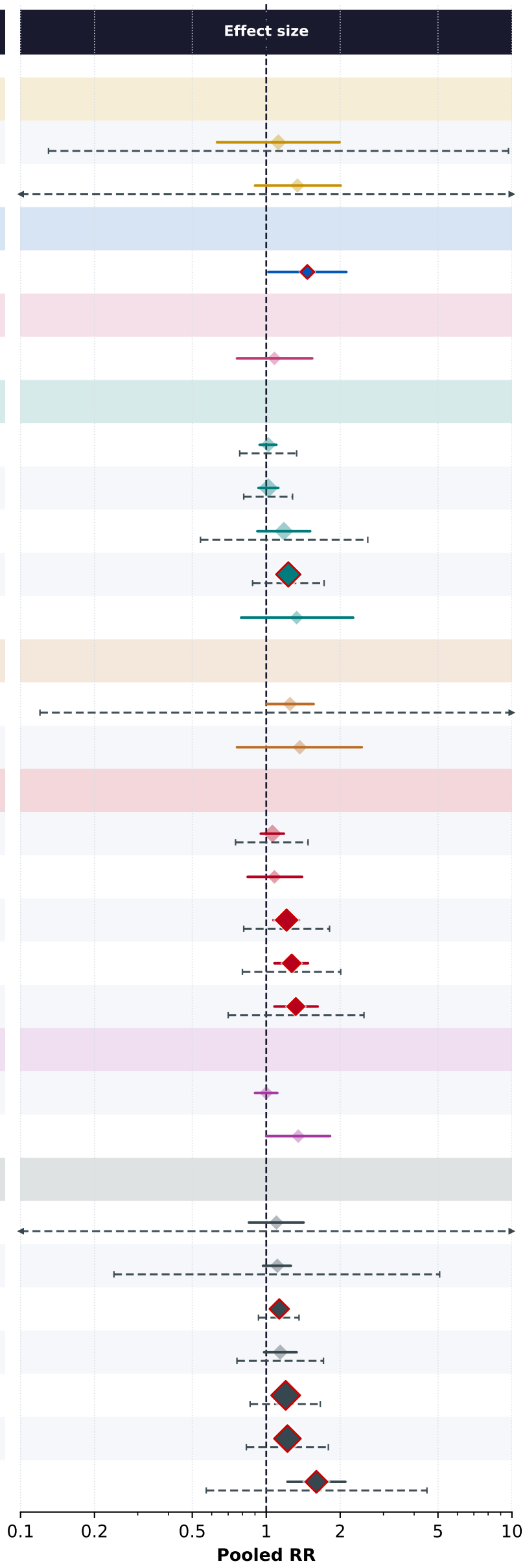
Exposures positively associated with prostate cancer risk
Meta-analysis of dietary-intake studies

Exposure	Studies	N	Cases	Pooled RR (95% CI)
VITAMINS A, C, D, E, K				
Vitamin A	5	3,207	1,358	*1.12 (0.63-1.99)*
Vitamin D	3	247,791	29,373	*1.34 (0.90-2.01)*
B VITAMINS				
Vitamin B12	2	719	245	*1.47 (1.02-2.12)*
ANTIOXIDANTS				
Antioxidants	2	50,507	7,155	*1.08 (0.76-1.54)*
MINERALS & TRACE ELEMENTS				
Iron	4	183,783	11,462	*1.02 (0.94-1.10)*
Selenium	10	235,441	18,652	*1.02 (0.93-1.12)*
Zinc	9	94,876	7,312	*1.18 (0.92-1.51)*
Calcium	19	770,289	-	*1.23 (1.12-1.36)*
Mineral supplements	2	57,251	1,958	*1.33 (0.79-2.26)*
FERMENTED FOODS & PROBIOTICS				
Yogurt	3	201,324	7,680	*1.25 (1.00-1.56)*
Fermented foods	3	3,460	339	*1.37 (0.76-2.45)*
FATTY ACIDS & LIPIDS				
Omega-3 fatty acids	8	607,190	20,690	*1.06 (0.95-1.18)*
Omega-9 fatty acids	2	200,518	7,678	*1.08 (0.84-1.40)*
Polyunsaturated Fat (PUFA)	13	673,771	-	*1.21 (1.07-1.36)*
Monounsaturated Fat (MUFA)	7	255,429	-	*1.27 (1.08-1.48)*
Alpha-linolenic acid	6	556,471	12,655	*1.32 (1.08-1.62)*
DIET				
Mediterranean diet	2	106,146	7,476	*1.00 (0.90-1.11)*
Western pattern diet	2	44,462	1,702	*1.35 (1.00-1.82)*
OTHER				
Ultra-processed food intake	3	4,983	2,932	*1.10 (0.85-1.42)*
Whole-grain intake	3	79,458	7,209	*1.11 (0.97-1.26)*
Processed meat intake	10	1,091,655	65,703	*1.13 (1.05-1.21)*
Multivitamin	4	348,954	4,443	*1.14 (0.98-1.33)*
Alcohol	29	1,934,957	-	*1.20 (1.10-1.30)*
Dairy	24	739,672	33,998	*1.22 (1.12-1.34)*
Red meat	14	974,550	57,630	*1.60 (1.22-2.10)*



Exposure group
Vitamins A, C, D, E, K
B Vitamins
Antioxidants
Minerals & Trace Elements
Fermented Foods & Probiotics
Fatty Acids & Lipids
Diet
Other

RR > 1 = positively associated with prostate cancer risk. |◆ pooled RR (size reflects number of studies) |◇ red outline = statistically significant (95% CI excludes 1.0) |◆ faded = not statistically significant | dashed line below CI = 95% prediction interval (where available) | arrowhead = interval continues beyond the axis

Include subexposures: Yes